DevOps

Kubernetes Services

iSTAD

Kubernetes Services

- Kubernetes Services provide a way to expose your application running in a Kubernetes cluster to other applications or users outside the cluster. A Service is an abstraction layer that provides a static IP address and a DNS name for a set of Pods. This allows other applications or users to access your application without having to know the IP address or location of the Pods running your application.
- In Kubernetes, there are four types of Services that you can use to expose your application: **ClusterIP**, **NodePort**, **LoadBalancer**, and **ExternalName**.

Deployment for Services

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deploy
spec:
 replicas: 3
 selector:
   matchLabels:
     app: nginx-app
 template:
   metadata:
     labels:
       app: nginx-app
   spec:
     containers:
       - name: nginx-cont
         image: nginx:1.22.1
         ports:
           - containerPort: 80
```

ISTAD

ClusterIP

- ClusterIP: This is the default Service type in Kubernetes. It provides a stable IP address and DNS name for your application within the cluster. This type of Service is ideal for communication between different parts of your application.

```yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-svc
spec:
  type: ClusterIP
  selector:
    app: nginx-app
  ports:
  - name: nginx-app-port
    port: 80
    targetPort: 80
```

ISTAD

NodePort

- NodePort: This type of Service exposes your application on a static port on each node in the cluster. This makes your application accessible from outside the cluster.

```yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-svc
spec:
  type: NodePort
  selector:
    app: nginx-app
  ports:
  - name: nginx-app-port
    port: 80
    targetPort: 80
    nodePort: 8181
```

LoadBalancer

- LoadBalancer: This type of Service provides a load balancer for your application. This makes your application accessible from outside the cluster and distributes incoming traffic to multiple Pods.

```yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-svc
spec:
  type: LoadBalancer
  selector:
    app: nginx-app
  ports:
  - name: nginx-app-port
    port: 80
    targetPort: 80
    nodePort: 8181
```

ISTAD

ExternalName

- ExternalName: This type of Service provides a DNS name for your application that resolves to an external address. This is useful when your application needs to communicate with an external service that doesn't have a Kubernetes Service.

```yaml
apiVersion: v1
kind: Service
metadata:
  name: my-external-svc
spec:
  type: ExternalName
  externalName: registry.e-crops.co
```

ExternalName

```yaml
apiVersion: v1
kind: Pod
metadata:
 name: dnsutils
 namespace: default
spec:
 containers:
 - name: dnsutils
   image: registry.k8s.io/e2e-test-images/jessie-dnsutils:1.3
   command:
     - sleep
     - "infinity"
   imagePullPolicy: IfNotPresent
 restartPolicy: Always
```

```
$ kubectl exec -i -t dnsutils -- nslookup my-external-svc
```

ExternalName

- `Call “my-external-svc” directly in Code`

```
const axios = require('axios');

async function makeExternalAPICall() {
 try {
   const response = await axios.get('http://my-external-svc/api/endpoint');
   console.log('Response:', response.data);
   // Process the response as needed
 } catch (error) {
   console.error('Error:', error.message);
   // Handle the error
 }
}

makeExternalAPICall();
```

Taint Node

- To prevent the master node in a Kubernetes cluster from running any services

```
$ kubectl taint nodes <master-node-name> node-role.kubernetes.io/master=:NoSchedule
```

- If you later decide to allow pods to run on the master node, you can remove the taint using the following command:

```
$ kubectl taint nodes <master-node-name> node-role.kubernetes.io/master-
$ kubectl get node # Get nodes name
```

```
$ kubectl taint nodes node5 service=disabled:NoSchedule
```

*service=disabled (Any key=value pair that you want notice or comment)

```
$ kubectl taint node node5 service-
```

*service (Just provide key name with “ - ”)

iSTAD

Thank you

Begin with the theory, Start with the practical